

MATH 526 Lab 6 Fall 2006

Part I

We learn to plot 2D graphs in this part. First issue the command

```
>>help plot
```

to get the basic information on **plot**.

We can draw the graph of the sine function over the interval $[-4, 4]$ using a partition of mesh size 0.01 as follows:

```
>>x = -4: .01: 4;
```

This creates the vector $\mathbf{x} = [-4 \ -3.99 \ -3.98 \ \cdots \ 3.98 \ 3.99 \ 4]$ with 801 components.

```
>>y = sin(x);
```

This creates the vector $\mathbf{y}$ whose components are the sines of the corresponding components of $\mathbf{x}$.

```
>>plot(x,y)
```

The graph of the function $y = \sin x$ over the interval $[-4, 4]$ is displayed in a figure window.

To clear the figure window, type

```
>>clf
```

We now draw the graph of $y = e^{x^2}$ over the interval $[-1.5, 1.5]$ using a partition of mesh size 0.01.

```
>>clear x y
```

```
>>x = -1.5: .01: 1.5;
```

This creates the vector $\mathbf{x} = [-1.5 \ -1.49 \ -1.48 \ \cdots \ 1.48 \ 1.49 \ 1.5]$ with 301 components.

```
>>y = exp(x.^2);
```

$\mathbf{x}.^2$ is the vector whose components are the squares of the corresponding components of $\mathbf{x}$. (Use **help arith** to learn more about array multiplication and array power.) The command **exp(x.^2)** then creates the vector $\mathbf{y}$ whose components are the exponentials of the corresponding components of the vector $\mathbf{x}.^2$. The net effect is that the components of $\mathbf{y}$ are the exponentials of the squares of the corresponding components of $\mathbf{x}$.

```
>>plot(x,y)
```

The graph of the function $y = \exp(x^2)$ over the interval $[-1.5, 1.5]$ is now displayed in the figure window.

We can add labels and title to the figure as follows:

```
>>xlabel('x-axis')
```

```
>>ylabel('y-axis')
```

```
>>title('My second plot')
```

Of course you can also check out **help xlabel**, **help ylabel** and **help title**.

We can also change the fill of the line used. For example, see what the following commands do:

```
>>clf
>>plot(x,y,'-.'
```

Create two other vectors **s** and **t** by the following commands:

```
>>t=-1.5:.1:1.5
```

The vector **t** contains the points that divide the interval $[-1.5, 1.5]$ into subintervals of length 0.1.

```
>>s=t.^3;
```

The components of the vector **s** are the cubes of the corresponding components of **t**.

Now plot the graph of the cubic polynomial $s = t^3$ over the interval $[-1.5, 1.5]$ by the command

```
>>plot(t,s)
```

The graph of $y = \exp(x^2)$ in the figure window is now replaced by the graph of $s = t^3$.

We can add the graph of $y = \exp(x^2)$ back to the figure window by typing the following commands.

```
>>hold on
```

The graph of $s = t^3$ is now held in the figure window.

```
>>plot(x,y)
```

Both graphs now appear in the figure window. Use **help hold** to learn more about the command **hold**.

Another way of putting two (or more) graphs in the same figure is to plot them simultaneously. Type the following commands:

```
>>figure
```

A new figure window is created.

```
>>plot(x,exp(x.^2),'r',x,x.^3,'b')
```

The graph of $y = \exp(x^2)$ now appears as a red curve in the new figure window and the graph of $y = x^3$ appears as a blue curve.

To close the figure windows, type

```
>>close all
```

Clear all variables before you work on Part II.